

Difference in Differences

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- Difference in means is biased due to the systematic differences between treatment and control groups

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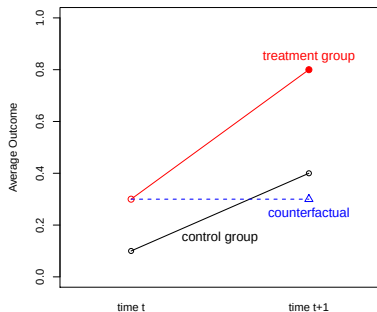
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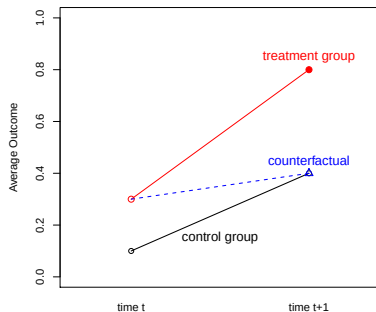
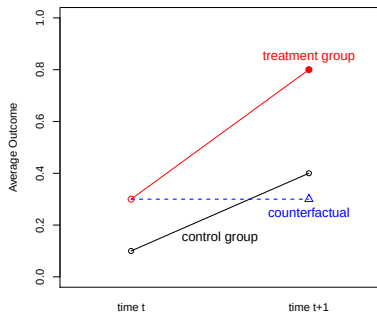
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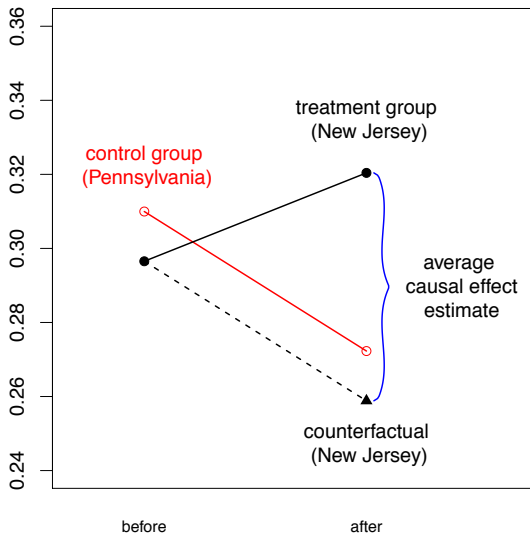
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- In 1992, NJ minimum wage increased from \$4.25 to \$5.05
 - Neighboring PA stays at \$4.25
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- NJ and (eastern) PA are similar
- Fast food chains in NJ and PA are similar: price, wages, products, etc.

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 - G_i : treatment ($G_i = 1$) or control ($G_i = 0$) group
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- DiD estimation:

$$\begin{aligned} \tau_{\text{DiD}} = & \underbrace{\{\mathbb{E}(Y_{i1} \mid G_i = 1) - \mathbb{E}(Y_{i0} \mid G_i = 1)\}}_{\text{difference for treated}} \\ & - \underbrace{\{\mathbb{E}(Y_{i1} \mid G_i = 0) - \mathbb{E}(Y_{i0} \mid G_i = 0)\}}_{\text{difference for control}} \end{aligned}$$

Regression perspective

- Time-fixed effect and individual-fixed effect are additive

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- Two-way fixed effects model:

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- $\tau_{\text{DiD}} = \tau$
- Least squares estimator equals the nonparametric DiD estimator
- The model can be applied for more general settings, but equivalence relationship **does not** hold in general beyond the 2×2 case

Comparison with the lagged outcome model

- Nonparametric identification assumption:

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- Neither stronger nor weaker than the parallel trend assumption
- Relationship between two methods: Ding and Li. 2019. *Political Anal.* “A bracketing relationship between difference-in-differences and lagged-dependent-variable adjustment”

Testing parallel trend

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- Modeling approach: suppose there are two pre-treatment periods and three post-treatment periods

$$Y_{it} = \alpha_i + \lambda_t + \delta_{-2}D_{it}^{-2} + \delta_{-1}D_{it}^{-1} + \delta_0D_{it}^0 + \delta_1D_{it}^1 + \delta_2D_{it}^2 + \epsilon_{it}$$

- for control group, all $D_{it} = 0$; for treatment group $D_{it}^k = 1$ if $t = k + 1$
- testing whether δ_{-2} , δ_{-1} are different from zero

Adjusting for Baseline Covariates

- Parallel trend assumption conditional on the baseline covariates:

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- Weighting (Abadie. 2005. *Rev. Econ. Stud*):

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where $\Pr(G_i = 1 \mid \mathbf{X}_i)$ is the propensity score

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- Unconditional parallel trend assumption neither implies nor is implied by conditional parallel trend assumption

Summary

- DID utilizes parallel trend assumption to identify the ATT
 - DID rests entirely on the parallel trend
 - parallel trend is sensitive to transformation of the outcome
- Generalization of parallel trend assumption
 - parallel trend on different scales
 - non-linear DID
- Generalization to multiple groups and multiple time periods

Suggested readings

- Classic DID
 - Card and Krueger. 1994 “Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania”
 - ANGRIST AND PISCHKE, Chapters 5
- Nonlinear DID
 - Athey and Imbens. 2006 “ Identification and inference in nonlinear difference-in-differences models”
 - Roth and Sant’ Anna. 2023 “When is parallel trends sensitive to functional form?”
- Multiple groups and multiple periods
 - Callaway and Sant’ Anna. 2021 “Difference-in-differences with multiple time periods”
 - de Chaisemartin and d’Haultfoeuille X. 2020 “Two-way fixed effects regressions with several treatments”
 - Sun and Abraham. 2021 “Estimating dynamic treatment effects in event studies with heterogeneous treatment effects”